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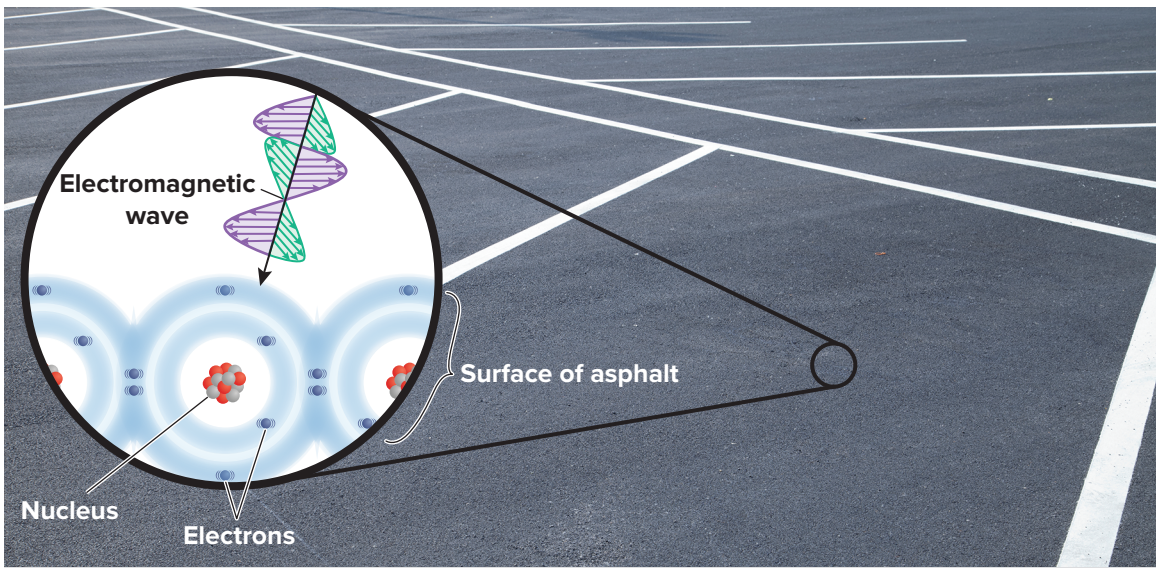


Figure 5 As an electromagnetic wave from the Sun strikes asphalt, charged particles in the asphalt gain energy from the vibrating electric and magnetic fields. This transfer of energy is why asphalt gets hot on sunny summer days.

Electromagnetic waves interact with matter

As an electromagnetic wave moves, it encounters objects. The vibrating electric and magnetic fields of the wave exert forces on the charged particles and magnetic materials that make up the object. This interaction causes the particles in the object to gain energy. For example, electromagnetic waves from the Sun cause electrons in asphalt to vibrate and gain energy, as shown in **Figure 5.** This energy can make the asphalt hot. The energy carried by an electromagnetic wave is called radiant energy. Radiant energy makes fire feel warm and enables you to see.

Waves and Particles

The difference between a wave and a particle might seem obvious—a wave is a disturbance that carries energy, and a particle is a piece of matter. However, in reality, the difference is not so clear.

Waves as particles

In 1887, Heinrich Hertz found that he could create a spark by shining light on a metal. (Today, we know that this spark means that electrons were ejected from the metal.) Hertz found that whether sparks occurred depended on the frequency of the light and not the amplitude. Because the energy carried by a sound wave or water wave depends on its amplitude and not its frequency, this result was mysterious.

Get It?

Identify What determines whether sparks are ejected from a metal when light shines on it?

In 1905, Albert Einstein provided an explanation. An electromagnetic wave can behave and can be modeled as a particle called a photon. A **photon** is a massless bundle of energy that behaves like a particle. The photon's energy depends on the frequency of the wave and increases as the wave's frequency increases. Although many features of electromagnetic waves are explained well by using the wave model, others are better explained using the particle model.

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